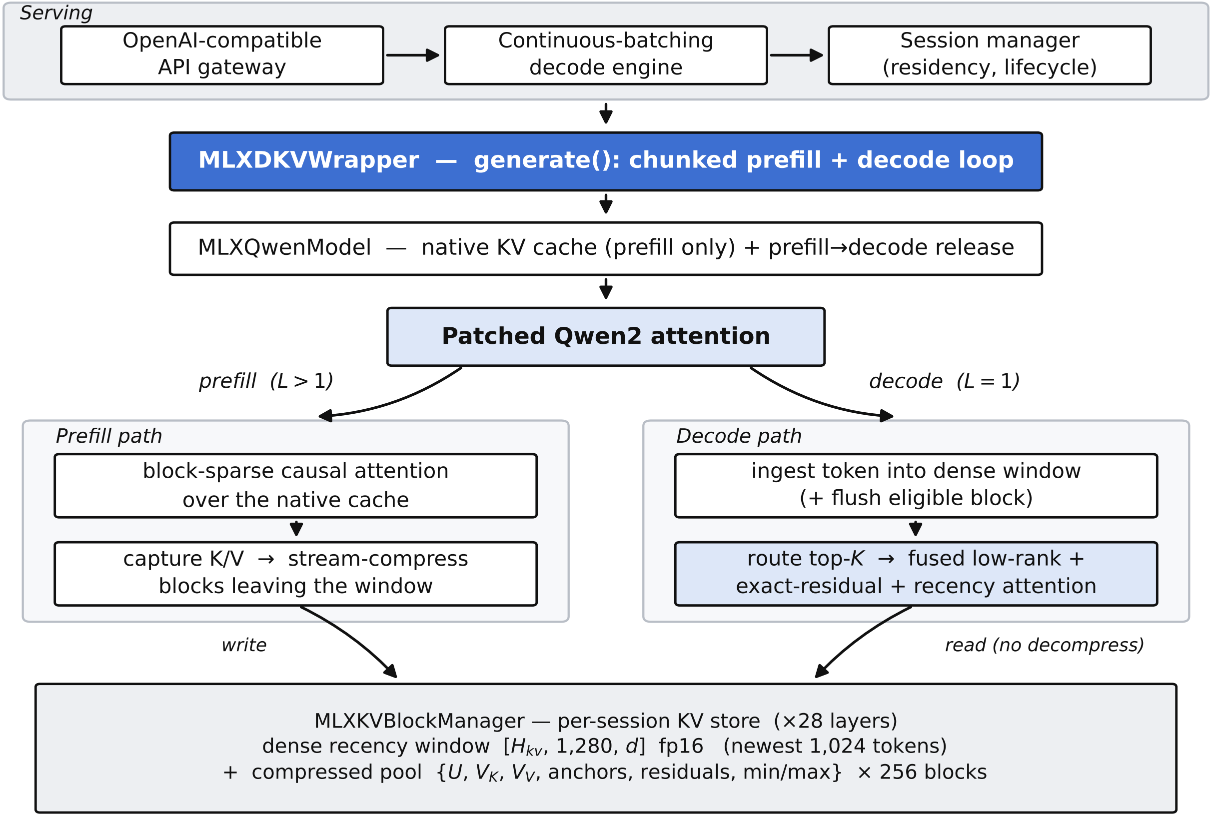
Serving
OpenAI-compatible API gateway
Continuous-batching decode engine
Session manager (residency, lifecycle)
MLXDKVWrapper — generate(): chunked prefill + decode loop
MLXQwenModel — native KV cache (prefill only) + prefill→decode release
Patched Qwen2 attention
prefill (L > 1)
decode (L = 1)
Prefill path
block-sparse causal attention over the native cache
capture K/V → stream-compress blocks leaving the window
Decode path
ingest token into dense window (+ flush eligible block)
route top-K → fused low-rank + exact-residual + recency attention
write
read (no decompress)
MLXKVBlockManager — per-session KV store (×28 layers)
dense recency window [H_{kv}, 1,280, d] fp16 (newest 1,024 tokens)
+ compressed pool {U, V_K, V_V, anchors, residuals, min/max} × 256 blocks